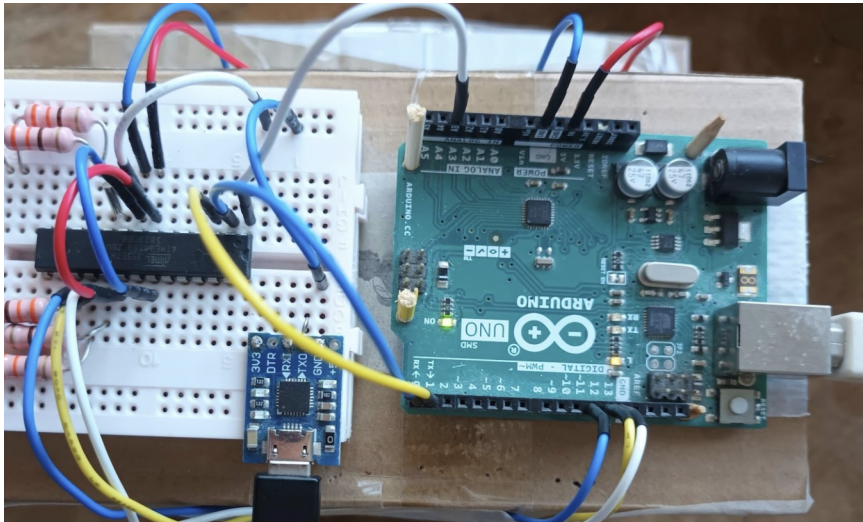


Installing a PCB111000 bootloader on an Atmega 168/3218

1. Loading an Atmega168 with the PCB111000_breadboard_V2 bootloader

Plug the Atmega168 into some plug-in breadboard and connect to an Arduino UNO as shown in fig 1 (ignore the resistors). Then follow the steps listed below the figure.



UNO pins	Atmega pins
5V	7 & 20
0V	8 & 22
13	19
12	18
11	17
A3	1
Rx	2
Tx	3
Watch crystal	9 & 10
CP2102 Module	Not connected

Fig 1 Using a UNO to calibrate, program and test an Atmega 168
Note the blue module: This is a CP2102 USB bridge It's use is optional

1. Open Arduino open "UNO_programs_168" and set the device to the UNO
Connect the UNO to the PC, temporarily disconnect the Rx & Tx wires and upload the sketch then reconnect the wires.
2. Open [Br@y++](#) and set the baud rate to 38400. Press -s- at the user prompt. Check that the Atmega168 has been detected and press -p-. Upload "Auto_cal_168" and follow the instructions to calibrate the device.
3. Press the reset button on the UNO and check calibration by uploading "Manual_Cal_168".
4. Reset the UNO again. This time press -P- and upload
"Bootloader_for_168\4_Full_bootloader.hex"
5. Set the Baud rate to 57600 and press any key when requested. The bootloader will respond with the user prompt "p/r p/r".
6. Press -p- and send a hex file that simply responds by sending data to the PC such as
"8_Integer_numbers.hex"
7. To try another program disconnect the UNO pin A3 and use the wire to reset the Atmega168 to restore the user prompt.
8. The bootloader can be made to run immediately from the -s- prompt (with A3 reconnected) by pressing -r-.

9. Note: The target program can not be made to run immediately following a POR with this setup. To do this upload “UNO_as_USB_bridge” to the UNO to configure it USB bridge. Disconnect A3. The UNO now replaces the CP2102 module and the user project will run immediately following a power cycle.

10. Disconnect the UNO entirely. The CP2102 module can now be connected as follows:
GND to Atmega pins 8 and 22
+3V3 to Atmega pins 7 and 20
TXO to Atmega pin 2 Rx
RXI to Atmega pin 3 Tx

Use Atmega pin 1 (reset) to upload a new user project.

2. Loading an Atmega 328 with the PCB111000_breadboard_bootloader

The procedure closely follows that adopted for the Atmega 168. However

1. The Arduino project “UNO_programs_168” is replaced by “UNO_programs_328”
2. The calibration routines are replaced by “Auto_cal_328” and “Manual_Cal_328”
3. The bootloader is replaced by Bootloader_for_328.hex

3. Building a PCB111000_Breadboard module

Follow the wiring instructions given in document

“PCB111000_Breadboard and Breadboard_V2 Build”.

The photo in this document shows a module with an 8 digit display. If a 4 digit display is required the right hand (RH) display module is used which contains the 4 lower significant bits.

4 Projects

A full set of projects is presented for both 4 digit and 8 digit displays for

“PCB111000_breadboard_V2”

that uses the Atmega168 bootloader.

A limited set is presented for “PCB111000_breadboard” that uses the Atmega328 bootloader. The effect of using the 328 is that its EEPROM contains 0x400 bytes rather than 0x200. It can be seen that 0x200 has been added to several EEPROM locations in particular those used to store the Clock calibration bytes.

It was also noted that accuracy of the 24 hour clock was poor when driven from an Atmega 328. It is assumed that the use of plug in breadboard and flying leads does not provide a sufficiently good ground for the crystal oscillator provided with the Atmega 328.